Freight Rate Prediction

Machine Learning Engineer Assessment | Approach, validation and final predictions

1. Objective and data

Predict the posted freight rate for 12,000 shipments dated November–December 2025. The available development dataset contains 48,000 labeled shipments from January–October 2025. Each row includes the route, mileage, equipment, weight, date and market-related features. The 31-row December chart uses a fixed Lexington-to-Fort Wayne shipment with only the date changing.

2. Data checks and preparation

Finding	Treatment and reason
292 negative weights	Used absolute values: their magnitudes resemble ordinary shipment weights.
300 missing training weights	Filled with the median for that equipment type, calculated using the training partition only.
374 missing training market indices	Used the same-day average market index, calculated from the feature data for that period.
Unusually high/low posted rates	Kept for evaluation. The regression stage reduces their influence during training by excluding extreme log-residual rows.

The model also uses log distance, log weight, geographic coordinates, an approximate road-to-straight-line distance ratio, weekday information and two alternatives for representing time trends. All engineered features are calculated without the target of the prediction period.

3. Model design

Stage 1 — Robust log-linear regression. A linear model predicts log(posted rate) from shipment details, market index and calendar features. After the initial fit, rows with absolute log residual above 0.25 are excluded from further regression fits. These rows are *not* removed from validation. Two time-trend specifications (quadratic and annual-seasonal) are averaged to reduce reliance on a single extrapolation assumption.

Stage 2 — LightGBM correction. LightGBM predicts the remaining log-rate error from shipment and route characteristics. It does not receive the continuous date trend, so it focuses on static pricing differences. Three fitted LightGBM models with different seeds are averaged.

Stage 3 — Lane correction. A small adjustment is learned for recurring pickup–delivery pairs from out-of-fold residual errors. A shrinkage term reduces unstable corrections, and unseen lanes receive no adjustment.

4. Validation approach

We used three chronological validation scenarios. Each model was trained on earlier months and evaluated on later months, matching the actual task better than random row splitting. All reported MAE values include the extreme-rate observations. Training-fold statistics, including missing-weight replacements, were recalculated for each fold.

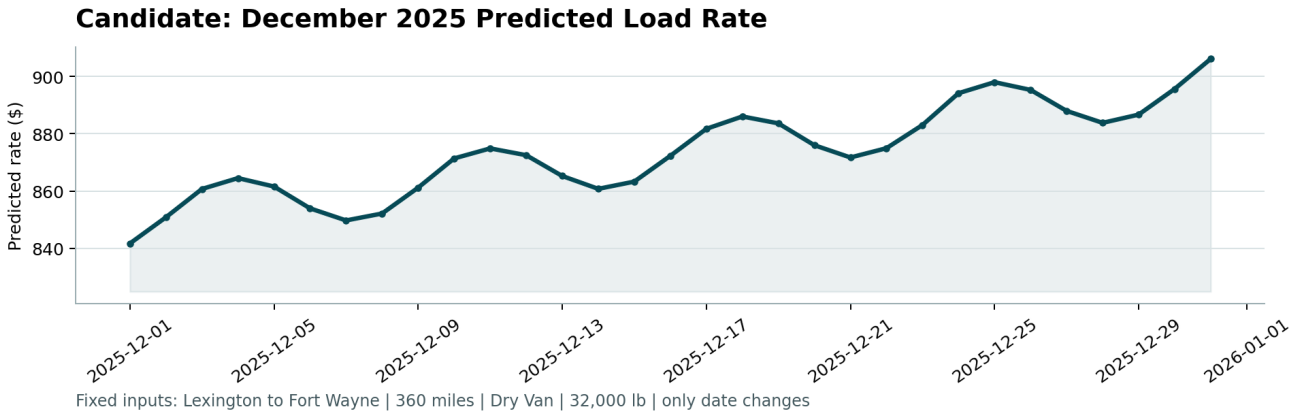
Training → validation	Robust regression	+ LightGBM	Full model
Jan–Jun → Jul–Aug	$111.46	$92.37	$89.97
Jan–Aug → Sep–Oct	$111.98	$89.54	$86.44
Jan–Sep → Oct	$116.87	$92.93	$89.50

Main result: The full model achieved an MAE of **$86.44** for September–October, compared with $111.98 for robust regression alone. That is a reduction of approximately 23%. The three final-model validation scores ranged from $86.44 to $89.97, supporting reasonably stable performance across the tested periods.

These are local historical-validation results, not the hidden score for the 12,000 final shipments. The final scorer checks output format and produces the December chart; Spotter evaluates final prediction accuracy separately. We tested many ideas during development, so these folds should not be treated as completely independent final-test estimates.

5. Final predictions and December chart

After validation, the selected model was refitted on all 48,000 labeled rows and used to predict the 12,000 unlabeled shipments. The required `validation_predictions.csv` contains only `load_id,predicted_rate`. The December chart holds pickup, delivery, distance (360 miles), equipment (Dry Van) and weight (32,000 lb) fixed while changing the date from December 1 to December 31.



December market inputs were obtained from the supplied unlabeled November–December shipments. The approach assumes those same-day feature values are available when making the predictions. The plotted daily curve is a model forecast, not observed December freight prices.

6. Limitations

Price spikes may represent real shipments or label errors; excluding them from regression fitting is a deliberate robustness choice. Only ten months of labeled data are available, making November–December trend extrapolation uncertain. Some cities appear only in the prediction dataset, so geographic generalization remains a risk. Market-index and quote-signal meanings are not explicitly defined in the assessment.